

Data Cleaning

A study was conducted, in which participants, between 18 and 99 years old, were asked to complete a task. Each participant received a score between 0 and 100. The scores that the participants achieved in the task were recorded, as well as their first names, last names, ages, genders, and amount of prior experience. Approximately 10,000 people participated in the study.

Unfortunately, a few mistakes were made in recording the data:

- Some participants' information was recorded in the database multiple times.
- Some participants did not specify their gender leaving it blank instead of as 'Male', 'Female', or 'Other'.
- Some participants specified their prior experience on a scale of 1 to 10 instead of 1 to 5 as instructed.
- For some participants, their scores were corrupted and replaced with NaN.

In this assignment, you will practice data cleaning. The input is a file named 'input.csv' (see reading materials), which will be read and saved as a table variable 'T'. You will write a function called 'clean_table'. This function will take the table variable 'T' as input and handle the following data cleaning tasks:

- Remove duplicate rows.
- Replace NaN values in the score column with the mean score of other participants.
- Replace any empty string in the gender column with 'unspecified'.
- For participants who entered prior experience using a 1-to-10 scale instead of the required 1-to-5 scale, divide their experience by 2 and round it down using the floor function.

Function

 Save

 Reset

 MATLAB Documentation (https://www.mathworks.com/help/)

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```
13 for i = nan_idx
14     T2.Score(i) = mu;
15 end
16
17 % You may use the fillmissing function. Here it is. You may explain it.
18 T2 = fillmissing(T2, 'constant', 'unspecified', 'DataVariables', {'Gender'});
19
20 % Access those with prior experience value >5 and floor them. You may use a similar way compared to the warm-up exercise.
21 % Your code here:
22 for i = 1:length(T2.Experience)
23     if T2.Experience(i) > 5
24         T2.Experience(i) = floor(T2.Experience(i)/2);
25     end
26 end
27
28 end
29
```

Code to call your function

 Reset

```
1 T = readtable("input.csv")
2 clean_table(T);
```

 Run Function

Output

T =

10010x7 table

First_Name	Last_Name	ID	Age	Gender	Experience	Score
{'Pvaddhjt' }	{'Psuwrah' }	13623	73	{'Male' }	3	44.654
{'Sexgyymbg' }	{'Bbnvoajw' }	14781	51	{'Male' }	1	50.891
{'Rfzn' }	{'Urhhq' }	2326	85	{'Female' }	2	56.487
{'Jutq' }	{'Sxlad' }	9484	98	{'Male' }	3	59.363
{'Fppasdy' }	{'Nnysez' }	17753	78	{'Male' }	7	56.894
{'Tttoh' }	{'Kqsgq' }	16922	21	{'Female' }	3	56.029
{'Bjza' }	{'Xhsuoczk' }	14268	52	{'Male' }	0	60.015
{'Uxdlsx' }	{'Zdhq' }	10938	58	{'Female' }	0	67.647
{'Caaevfgi' }	{'Mthat' }	16617	57	{'Male' }	3	48.008
{'Rpejkybb' }	{'Fudgofgir' }	12732	42	{'Female' }	1	54.091
{'Zdglx' }	{'Nbpzzkkso' }	8157	53	{'Female' }	2	63.916
{'Aodvmz' }	{'Amymvag' }	17537	64	{'Female' }	2	58.436
{'Ulelep' }	{'Vhrdzrd' }	5563	37	{'Male' }	0	64.959
{'Dmeuiopu' }	{'Goyfnvq' }	10235	26	{'Male' }	2	55.963
{'Xpnnvqr' }	{'Dfsbyxunf' }	5579	39	{'Female' }	3	56.695
{'Jdafarsec' }	{'Cfuw' }	4107	25	{'Male' }	0	74.009
{'Dckn' }	{'Drbxddp' }	9208	63	{'Male' }	3	76.24
:	:	:	:	:	:	:
{'Baztqdnlb' }	{'Mmzxvxq' }	14733	38	{'Male' }	5	49.702
{'Zxqpipyni' }	{'Xwoulv' }	6324	28	{'Female' }	1	90.117
{'Wukfq' }	{'Evqjvpd' }	9138	87	{'Female' }	2	38.967
{'Ntzmsi' }	{'Dkdtm' }	18127	43	{'Male' }	2	52.771
{'Ysmm' }	{'Diaqs' }	18526	57	{'Male' }	2	76.878
{'Uf1ycrvyq' }	{'Qwleks' }	11026	56	{'Male' }	2	69.399
{'Jczahhwg' }	{'Pbfxngu' }	13642	26	{0x0 char}	3	64.381
{'Xbkifz' }	{'Pvjvhupgg' }	17108	54	{'Other' }	3	41.047
{'Iqwozcp' }	{'Rfwdx' }	4229	44	{'Male' }	4	68.421
{'Niphesbp' }	{'Lszpzcjff' }	4298	80	{'Female' }	1	31.763
{'Irtuu' }	{'Jbdrseaj' }	16938	78	{'Female' }	4	53.134
{'Emribj' }	{'V1fzmfhfh' }	13453	26	{'Female' }	1	42.845
{'Ulibdycw' }	{'Aiatm' }	7832	86	{'Male' }	4	30.132
{'Mjxxhpoj' }	{'Ivjjdffvp' }	16751	51	{'Female' }	2	41.706
{'Dzjff' }	{'Gtkcw' }	1678	78	{0x0 char}	8	56.807
{'Zbmuljc' }	{'Enhs' }	12713	48	{'Male' }	3	47.889
{'Tzajilc' }	{'Wtgnpvssi' }	9553	87	{'Male' }	3	68.135

Display all 10010 rows

Tests: All Tests Passed

Submit

✔ Removing duplicate rows

✔ Replacing empty names

✔ Replace NaNs

✔ Maximum experience is 5
✔ Checking entire table